Howard Tang

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EDUCATION

Princeton University Princeton, NJ · Expected Graduation: May 2029
B.S.E. in Electrical and Computer Engineering GPA: 3.78 / 4.00
Relevant Coursework: Data Structures and Algorithms, Introduction to Machine Learning, Contemporary Logic Design

RESEARCH EXPERIENCE

National Center for Supercomputing Applications (NCSA) · *Research Intern* Urbana-Champaign, IL · May – July 2026

- Built FO4DSMPLX, a modular computer vision framework for human-centric 4D scene reconstruction that integrates open-source perception models (SMPLest-X human mesh recovery, ViTPose++ pose estimation, VGGT-Ω 3D reconstruction, SAM 2 segmentation) with Umeyama + RANSAC alignment; evaluated on 7 in-the-wild scenes, presented at a research exposition, and open-sourced on GitHub.
- Cut end-to-end inference time by 50% on an NVIDIA A40 GPU; configured CUDA 12.4 toolchains and CUDA-enabled PyTorch3D builds on NCSA's Delta cluster.
- Automated parallel GPU experiments with SLURM on Delta HPC to stress-test two video-diffusion pipelines (FreeOrbit4D, Uni3C) on 30 adversarial inputs; found that lower input resolution improves temporal and geometric consistency in Uni3C, opening a new research direction.

ROBOTICS & AUTONOMOUS SYSTEMS

Semi-Autonomous Robotic Vehicle · *Designer & Builder* Rochester, NY · Sept. 2023 – March 2025

- Built a semi-autonomous differential-drive robot from scratch over 5 design cycles, with custom multi-layer PCBs (EasyEDA) and a hand-wired, soldered board (Teensy 4.0, voltage regulators, sensors).
- Wrote all embedded C/C++ firmware: ~175 Hz real-time control loop with motor control and sensor interfacing (ToF LiDAR, ultrasonic distance sensors, magnetic encoders, IMU).
- Achieved odometry-based localization within ~3 mm over 1.5 m and ~1 cm over 8 m by fusing IMU and encoder data (sensor fusion) with a complementary filter; fit skid-distance regression curves to improve obstacle detection.
- Placed 3rd among 70+ teams at the 2025 NY Science Olympiad Robot Tours state competition with autonomous runs.

Urban League of Rochester · *Engineering Intern* Rochester, NY · June – July 2025

- Assembled, debugged, and flight-tested 12 quadcopter drones (flight controllers, ESCs, RF links); wrote documentation enabling student operation.

DIGITAL HARDWARE DESIGN

16-Bit Processor & RTL Design in Verilog · *Contemporary Logic Design* Princeton, NJ · Fall 2025

- Designed a 16-bit, 16-instruction, Turing-complete microprocessor in Verilog from scratch (based on the LC-3 ISA), implementing the full fetch–decode–execute cycle with a control FSM, register file, datapath, and memory interface.
- Verified it with self-written modular testbenches achieving 100% instruction coverage (waveform debugging in Xilinx Vivado) and wrote a custom assembler to run user-defined programs on it.
- Completed 70+ hours of RTL labs (FSMs, HLSMs, datapaths) in Vivado and recreated the Simon game as a Verilog FSM.

SOFTWARE & MACHINE LEARNING PROJECTS

Course Projects · *Data Structures & Algorithms; Introduction to Machine Learning* Princeton, NJ · Fall 2025 – Present

- Programmed a fraud-detection pipeline (graph algorithms, MSTs, decision stumps, modified AdaBoost), plus a seam-carving image compressor (dynamic programming), a k-d tree, and an autocomplete engine.
- Trained classification and regression models in Python and implemented Word2Vec, PCA, and k-means clustering.

TECHNICAL SKILLS

Programming: Python, C/C++ (embedded), Java, Verilog
Robotics & Controls: PID control, sensor fusion, state estimation, odometry/localization, differential-drive kinematics
Computer Vision & ML: PyTorch, 3D reconstruction, pose/mesh estimation (SMPL-X), segmentation (SAM), diffusion models
Embedded, Hardware & HPC: Teensy/Arduino firmware, PCB design, RTL design (Vivado), SLURM, CUDA GPU workflows
CAD & Fabrication: Fusion 360 (100+ designs), Onshape, FDM 3D printing, laser cutting

ACTIVITIES

Member, Princeton University Robotics Club (PURC) · **Member**, Princeton Autonomous Vehicle Engineering (PAVE)